

RovoTools — Phase 0

Complete Architecture, SEO & Growth Audit

You are a senior full-stack architect, Next.js engineer, technical SEO engineer, performance engineer, security engineer and digital-growth architect.

You are working on the existing RovoTools website:

<https://rovotools.com/>

IMPORTANT:

This is an existing production project.

DO NOT modify, delete, rename or rewrite application files during this phase.

This phase is AUDIT ONLY.

Your first responsibility is to understand the existing codebase before recommending implementation changes.

1. Inspect the complete project

Inspect:

- package.json
- lock file
- Next.js configuration
- TypeScript configuration
- ESLint configuration
- Tailwind configuration
- shadcn/ui configuration
- app directory
- pages directory if present
- components
- lib
- utils
- hooks
- API routes
- database/schema
- public directory
- image assets

- fonts
- PWA configuration
- authentication if present
- analytics
- AdSense
- blog
- tool registry
- category registry

Determine the actual architecture.

Do not assume the architecture from the project description.

2. Create a complete route inventory

Find every public route.

Classify each route:

- Homepage
- Tool
- Category
- Blog
- Blog article
- Static information page
- Authentication
- Admin
- API
- Error
- Dynamic route
- Other

Produce a route table containing:

- URL
- route file
- page type
- indexable?
- canonical?

- metadata?
 - schema?
 - breadcrumbs?
 - internal links?
 - status
-

3. Create a tool inventory

Find every existing tool.

For each tool identify:

- tool name
- slug
- route
- category
- component
- description
- SEO title
- SEO description
- H1
- FAQ
- related tools
- related guides
- processing model
- client-side/server-side processing
- analytics
- share functionality
- download functionality
- copy functionality

Identify:

- duplicate tools
- orphan tools
- broken tools
- missing metadata

- missing categories
 - missing routes
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4. Audit technical SEO

Inspect:

- title tags
- meta descriptions
- canonical URLs
- robots metadata
- robots.txt
- sitemap.xml
- Open Graph
- Twitter/X metadata
- favicon
- WebSite schema
- Organization schema
- WebApplication/software schema where used
- Breadcrumb schema
- Article schema
- FAQ schema
- ItemList schema

Identify:

- duplicate titles
 - duplicate descriptions
 - missing metadata
 - incorrect canonical URLs
 - noindex mistakes
 - indexable duplicate URLs
 - missing structured data
 - malformed structured data
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5. Audit internal linking

Determine whether the current structure is:

Homepage

→ Category

→ Tool

→ Related Tool

→ Guide

Identify:

- orphan pages
 - weak category pages
 - tools with no related tools
 - tools with no related guides
 - weak breadcrumbs
 - excessive links
 - broken internal links
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6. Audit content architecture

Inspect:

- homepage content
- category descriptions
- tool introductions
- how-to sections
- FAQs
- blog articles
- related articles
- tool/article connections

Identify pages that have useful content versus thin content.

Do NOT recommend adding word count merely for SEO.

7. Audit performance

Inspect:

- client components
- server components
- JavaScript bundle

- unnecessary dependencies
- image optimization
- font loading
- third-party scripts
- AdSense
- analytics
- hydration
- API calls
- unnecessary re-renders
- large assets

Identify likely:

- LCP problems
- INP problems
- CLS problems
- TTFB problems

Do not modify anything.

8. Audit accessibility

Check:

- semantic HTML
- headings
- labels
- keyboard navigation
- focus states
- buttons
- forms
- contrast
- ARIA
- mobile usability
- screen-reader issues

9. Audit security

Check architecture for:

- XSS
- unsafe HTML
- unsafe URL handling
- SSRF
- file uploads
- malicious SVG
- path traversal
- API abuse
- rate limiting
- exposed secrets
- client-side secrets
- dependency vulnerabilities

Pay special attention to:

- website analyzer
- SEO checker
- image tools
- PDF tools
- URL-based tools
- developer tools

Do not exploit anything.

Only identify risks.

10. Audit privacy claims

For every tool determine whether processing is:

- local/browser
- server
- third-party
- hybrid

Compare actual implementation with privacy messaging.

Identify any claim such as:

"Your files never leave your device"

that is not technically supported by the implementation.

Do not change claims during this phase.

11. Audit monetisation

Inspect:

- AdSense
- ad placement
- ad loading
- mobile ads
- tool/result positioning
- possible accidental ad/control confusion

Identify UX risks.

12. Audit growth architecture

Determine whether the platform currently supports:

- tool search
 - tool discovery
 - trending tools
 - popular tools
 - shareable results
 - generated URLs
 - social sharing
 - OG previews
 - backlinks
 - embeds
 - tool analytics
 - returning-user features
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13. Audit programmatic SEO potential

Identify legitimate opportunities for:

- calculator landing pages
- image-compression landing pages

- color conversion pages
- developer-tool pages
- SEO-tool pages
- PDF-tool pages

Do NOT recommend thousands of thin keyword pages.

Only identify pages where there is meaningful search intent and unique functionality.

14. Audit content gaps

Identify potential high-value missing tools in:

- Calculators
- Developer
- Image
- Design
- Color
- PDF
- SEO
- Finance
- Security
- Text
- QR/Barcode
- Student
- Marketing

Rank opportunities by:

- user usefulness
- search intent
- implementation complexity
- potential internal-link value

Do NOT provide fake search-volume numbers.

15. Create the audit report

Create:

/docs/ROVOTOOLS-BASELINE-AUDIT.md

The report must contain:

1. Executive Summary
2. Existing Architecture
3. Route Inventory
4. Tool Inventory
5. SEO Audit
6. Metadata Audit
7. Structured Data Audit
8. Internal Linking Audit
9. Content Audit
10. Performance Audit
11. Accessibility Audit
12. Security Audit
13. Privacy Audit
14. Monetisation Audit
15. Growth Audit
16. Programmatic SEO Opportunities
17. Missing Features
18. Technical Debt
19. Recommended Priority Order

For each problem include:

- Severity: Critical / High / Medium / Low
- Evidence
- File/path
- Recommended solution
- Estimated implementation complexity

16. Final response

DO NOT modify application code.

Report:

- number of routes
- number of tools

- number of categories
- number of blog articles
- SEO issues
- performance issues
- security issues
- privacy issues
- growth opportunities
- top 10 priorities

Then stop.

Do not proceed automatically to Phase 1.

Update — 2026-10-03: Phases, Skills, Baseline Audit, Code Fixes

This update splits the program into docs/phases/ (README + PHASE-0..PHASE-6), adds .opencode/skills/ (6 skills), records Phase 0 baseline audit results, and applies safe code fixes. Source of truth for phases: docs/phases/. Audit report: docs/ROVOTOOLS-BASELINE-AUDIT.md.

Phase files (docs/phases/)

- README.md — index, order, exit criteria
- PHASE-0-baseline-audit.md — audit-only, 19-section report
- PHASE-1-technical-seo-fixes.md — titles/meta/canonical/robots/sitemap/schema
- PHASE-2-content-linking.md — Home/Category/Tool/Guide, breadcrumbs, FAQs
- PHASE-3-performance-a11y.md — LCP/INP/CLS/TTFB + WCAG 2.2 AA
- PHASE-4-security-privacy.md — XSS/SSRF/uploads + claim alignment
- PHASE-5-monetisation-growth.md + PHASE-6-programmatic-gaps.md

Skills (.opencode/skills/)

- seo-audit — Phase 0 read-only workflow
- seo-fix — Phase 1 static-export fixes
- content-linking — Phase 2
- perf-a11y — Phase 3
- security-privacy — Phase 4
- growth-programmatic — Phases 5-6

Baseline audit results (verified 2026-10-03)

Routes: 16 page.tsx | Tools: 92 (88 catalog slugs + 4 seed) | Categories: 13 | Blog posts: 13 | Sitemap ~129 URLs (11 static + 13 cat + 92 tool + 13 blog) | SEO decks 34/92 | .htaccess 301s 16 | route.ts 0. Static export output:export, trailingSlash:true, images.unoptimized:true. Full 19-section report in docs/ROVOTOOLS-BASELINE-AUDIT.md.

Top 10 priorities

1. Verify youtube-thumbnail-downloader/network tools client-only + allowlist (Critical)
2. Privacy claim fixed: header absolute to nuanced (High) — DONE
3. Canonical trailing-slash alignment + sitemap timestamps (High)
4. Wire analytics provider or remove dead code (High)
5. 58 fallback decks expansion (Medium)

6. Per-tool OG + WebSite schema + Org logo (Medium) — WebSite/Org logo DONE
7. Image perf + PDF chunking + LHCI (Medium)
8. Tool-guide coverage (Medium)
9. Validator/formatter hubs, social/RSS/embeds (Low)
10. A11y manual + link-checker CI (Low)

Code changes applied

- apps/web/src/components/Header.tsx: banner updated to nuanced privacy claim (typecheck+lint clean).
- apps/web/src/app/layout.tsx: Organization JSON-LD + logo/alternateName, new WebSite JSON-LD. typecheck+lint clean.
- Docs: docs/phases/ (8 files), .opencode/skills/ (6 SKILL.md), docs/ROVOTOOLS-BASELINE-AUDIT.md (19 sections).

Updated: 2026-10-03T20:38:49Z